Majorana Modes in the Machine

Block 1: The Physics Bridge

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Outline

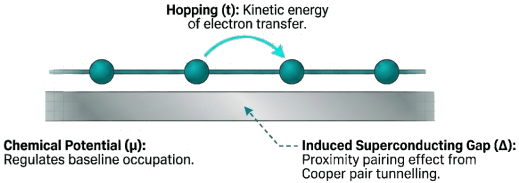
1. The Kitaev Hamiltonian
2. Majorana Representation
3. BdG Hamiltonian & Spectrum
4. Topological Invariant
5. Topology & Protection
6. Summary

Starting point: the lattice Kitaev Hamiltonian

$$H_K = -\mu \sum_j c_j^\dagger c_j - t \sum_j (c_j^\dagger c_{j+1} + \text{h.c.}) + \Delta \sum_j (c_j^\dagger c_{j+1}^\dagger + \text{h.c.}) \tag{1}$$

Physical ingredients

- Spinless (or spin-polarized) fermions
- Nearest-neighbor hopping t
- p-wave pairing $\Delta \in \mathbb{R}$
- Chemical potential μ tunes filling



Questions we will answer

- When does topology emerge?
- How do edge zero modes appear?
- What is the topological invariant?

What are Majorana fermions and why simulate them?

- The **Majorana operator** is Hermitian:

$$\gamma = \gamma^\dagger$$

- In the Kitaev chain, ordinary fermions split into two **Majorana modes**:

$$c_j = \frac{1}{\sqrt{2}}(\gamma_{A,j} + i\gamma_{B,j})$$

- In the **topological phase**, these Majoranas:
  - pair up *between sites*
  - leave **unpaired zero-energy modes at the edges**

Why is this interesting?

Edge Majorana modes are **robust** (protected by topology) and candidates for **fault-tolerant quantum computation**. Simulating the model helps understand topological phases and non-local quantum information storage.

Majorana operators

Define

$$\gamma_{A,j} = \frac{c_j + c_j^\dagger}{\sqrt{2}}, \qquad \gamma_{B,j} = \frac{i(c_j^\dagger - c_j)}{\sqrt{2}} \tag{2}$$

so that $\{\gamma_{\alpha,i}, \gamma_{\beta,j}\} = 2\delta_{\alpha\beta}\delta_{ij}$.

Rewriting the Hamiltonian:

$$H_K = -\mu \sum_j \left(i\gamma_{A,j}\gamma_{B,j} + \tfrac{1}{2} \right) + i \sum_j \left[(\Delta + t)\, \gamma_{B,j}\gamma_{A,j+1} + (\Delta - t)\, \gamma_{A,j}\gamma_{B,j+1} \right] \tag{3}$$

Sweet spot: $\mu = 0,\ t = \Delta$

Only inter-site coupling $i\gamma_{B,j}\gamma_{A,j+1}$ survives. Open boundaries leave **one unpaired Majorana at each end**.

Momentum Space Mapping: Fourier Transform

The tight-binding Hamiltonian in real space is defined as:

$$H = \sum_j \left[-t(c_j^\dagger c_{j+1} + \text{h.c.}) - \mu \left(c_j^\dagger c_j - \frac{1}{2} \right) + \Delta(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger) \right] \tag{4}$$

Applying the Fourier transform $c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikj} c_k$, the kinetic and chemical potential terms diagonalize trivially. The pairing term requires antisymmetrization due to fermionic statistics ($c_k c_{-k} = -c_{-k} c_k$):

$$\Delta \sum_k e^{ik} c_k c_{-k} = \Delta \sum_{k>0} (e^{ik} - e^{-ik}) c_k c_{-k} = 2i\Delta \sum_{k>0} \sin(k) c_k c_{-k} \tag{5}$$

This isolates the odd-parity p-wave superconducting nature of the model in momentum space.

BdG Formalism and Pseudospin Projection

Use the Nambu spinor $\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}$ to cast H into a bilinear form.:

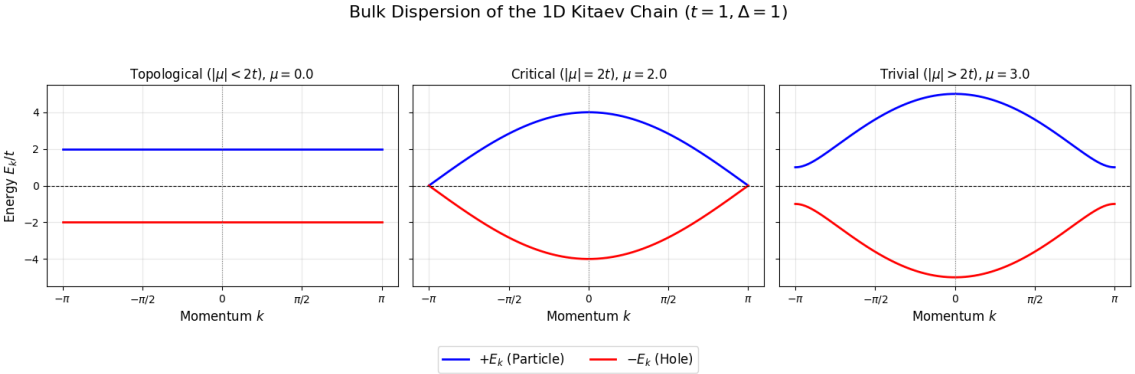
$$H = \sum_{k>0} \Psi_k^\dagger h(k) \Psi_k, \quad h(k) = \begin{pmatrix} \xi_k & 2i\Delta \sin(k) \\ -2i\Delta \sin(k) & -\xi_k \end{pmatrix} \tag{6}$$

normal-state dispersion $\xi_k = -\mu - 2t\cos(k)$.

$$h(k) = n_y(k)\sigma^y + n_z(k)\sigma^z \implies \begin{cases} n_z(k) = -\mu - 2t\cos(k) \\ n_y(k) = -2\Delta \sin(k) \end{cases} \tag{7}$$

$E_k = \pm\sqrt{n_z(k)^2 + n_y(k)^2}$. The topological phase transition : $E_k = 0$. This occurs at the points $k = 0, \pi$ where $n_y(k) \equiv 0$,condition $\mu = \pm 2t$.

Bulk dispersion: topological / critical / trivial

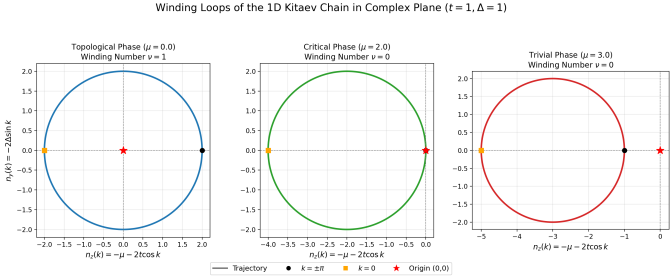


Gap closes at $k = 0$ when $\mu = -2t$ (and at $k = \pi$ when $\mu = +2t$).

Winding number ν

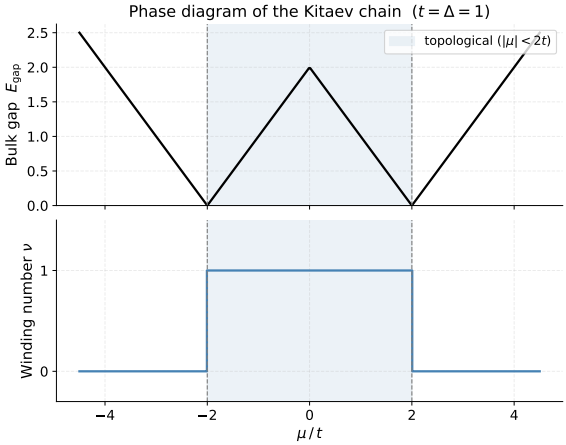
As k sweeps the BZ, the vector $\mathbf{n}(k) = (n_z, n_y)$ traces a closed loop.

$$\theta_k = \arg(n_z + i n_y), \qquad \nu = \frac{1}{2\pi} \int_{\mathrm{BZ}} \partial_k \theta_k \, dk \tag{8}$$



- $\nu = 0$: loop does *not* enclose origin $\rightarrow$ trivial
- $|\nu| = 1$: loop winds once around origin $\rightarrow$ topological

Phase diagram



Bulk gap vanishes at $\mu = \pm 2t$; winding number jumps across the transition.

Topology: Continuity and Invariance

- **Topological Equivalence**: Two shapes are equivalent if they can be transformed into each other via "continuous deformation" (stretching, bending).

Topology is "Rubber Geometry"

- **Invariants**: Properties that remain unchanged during deformation (e.g., number of holes / Genus).
- **Physical Link**: Tuning Hamiltonian parameters (t, μ) is essentially a "continuous deformation" of the wave function.

Core Intuition

You cannot turn a Sphere (0 holes) into a Donut (1 hole) without "tearing" the surface (Gap Closing).

Mapping to k-space: The Winding Number

The Kitaev Hamiltonian in d_z-d_y plane:

$$d_z(k) = -2t\cos k - \mu, \quad d_y(k) = 2\Delta \sin k$$

- **Trivial Phase ($|\mu| > 2t$)**: The trajectory does not enclose the origin, $\nu = 0$.
- **Topological Phase ($|\mu| < 2t$)**: The trajectory **winds around the origin** ($E = 0$ singularity), $\nu = 1$.

What is Protection?

As long as the circle doesn't cross the origin (Gap remains open), parameter fluctuations only "deform" the circle but cannot change the **integer** winding number.

The Consequence: Zero-Energy Edge States

- **Bulk-Boundary Correspondence**: A non-trivial winding ($\nu = 1$) in the bulk necessitates an edge state to compensate the discontinuity at the vacuum interface.
- **Inevitability**: This compensation appears as a state with **eigenvalue strictly at 0** (Majorana Zero Mode).
- **Spatial Profile**: Unlike bulk waves, these states are exponentially localized at the chain ends.

$$\textit{Edge "unpaired" sites} \implies \textit{Topologically protected zero modes}$$

Block 1 — key takeaways

1. The Kitaev Hamiltonian hosts **two phases** separated by gap closings at $\mu = \pm 2t$.
2. Majorana representation reveals the **sweet-spot** picture directly ($\mu = 0,\ t = \Delta$).
3. The BdG d-vector winding number $\nu \in \{0, 1\}$ is the **topological invariant**.

Next (Block 2): Finite size & edge modes $\rightarrow$ Jordan–Wigner transform $\rightarrow$ qubit encoding $\rightarrow$ quantum circuit for ground-state preparation.